Energy, Work & Power

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Work Done by Constant & Inclined Forces**

A constant force of 15 N is applied to an object, causing it to move a distance of 4 m in the direction of the force. Calculate the work done by the force.

Final Answer: 60 J. The work done by a constant force is given by the formula: Work Done = Force × Distance × cos(θ) where θ is the angle between the force and the direction of motion. In this case, the force and the distance are in the same direction, so θ = 0°. Therefore, cos(0°) = 1. Work Done = 15 N × 4 m × 1 Work Done = 60 J

2 **[Foundational]** • **Target: Kinetic Energy & Potential Energy**

A force of 42 N acts at 30° to the direction of motion of a body that moves 27 m. Find the work done, leaving your answer in exact (surd) form.

Final Answer: $1134 \times \frac{\sqrt{3}}{2}$ J. $W = Fd\cos 30^\circ = 42 \times 27 \times \frac{\sqrt{3}}{2} = 1134 \times \frac{\sqrt{3}}{2}$ J

3 **[Foundational]** • **Target: The Work-Energy Principle**

A force of 20 N acts at 45° to the direction of motion of a body that moves 10 m. Find the work done, leaving your answer in exact (surd) form.

Final Answer: $200 \times \frac{\sqrt{2}}{2}$ J. $W = Fd\cos 45^\circ = 20 \times 10 \times \frac{\sqrt{2}}{2} = 200 \times \frac{\sqrt{2}}{2}$ J

4 **[Foundational]** • **Target: Power and Vehicles in Motion**

A force of 28 N acts at 45° to the direction of motion of a body that moves 28 m. Find the work done, leaving your answer in exact (surd) form.

Final Answer: $784 \times \frac{\sqrt{2}}{2}$ J. $W = Fd\cos 45^\circ = 28 \times 28 \times \frac{\sqrt{2}}{2} = 784 \times \frac{\sqrt{2}}{2}$ J

5 **[Foundational]** • **Target: Work Done by Constant & Inclined Forces**

A force of 42 N acts at 45° to the direction of motion of a body that moves 14 m. Find the work done, leaving your answer in exact (surd) form.

Final Answer: $588 \times \frac{\sqrt{2}}{2}$ J. $W = Fd\cos 45^\circ = 42 \times 14 \times \frac{\sqrt{2}}{2} = 588 \times \frac{\sqrt{2}}{2}$ J

6 **[Foundational]** • **Target: Kinetic Energy & Potential Energy**

A force of 40 N acts at 30° to the direction of motion of a body that moves 15 m. Find the work done, leaving your answer in exact (surd) form.

Final Answer: $600 \times \frac{\sqrt{3}}{2}$ J. $W = Fd\cos 30^\circ = 40 \times 15 \times \frac{\sqrt{3}}{2} = 600 \times \frac{\sqrt{3}}{2}$ J

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7 [Foundational] • Target: The Work-Energy Principle

A force of 75 N acts at 45° to the direction of motion of a body that moves 8 m. Find the work done, leaving your answer in exact (surd) form.

Final Answer: $600 \times \frac{\sqrt{2}}{2}$ J. $W = Fd\cos 45^\circ = 75 \times 8 \times \frac{\sqrt{2}}{2} = 600 \times \frac{\sqrt{2}}{2}$ J

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Power and Vehicles in Motion

A force of 78 N acts at 45° to the direction of motion of a body that moves 4 m. Find the work done, leaving your answer in exact (surd) form.

Final Answer: $312 \times \frac{\sqrt{2}}{2}$ J. $W = Fd\cos 45^\circ = 78 \times 4 \times \frac{\sqrt{2}}{2} = 312 \times \frac{\sqrt{2}}{2}$ J

9 [Intermediate] • Target: Work Done by Constant & Inclined Forces

A force of 64 N acts at 60° to the direction of motion of a body that moves 19 m. Find the work done, leaving your answer in exact (surd) form.

Final Answer: $1216 \times \frac{1}{2}$ J. $W = Fd\cos 60^\circ = 64 \times 19 \times \frac{1}{2} = 1216 \times \frac{1}{2}$ J

10 [Intermediate] • Target: Kinetic Energy & Potential Energy

A force of 57 N acts at 60° to the direction of motion of a body that moves 21 m. Find the work done, leaving your answer in exact (surd) form.

Final Answer: $1197 \times \frac{1}{2}$ J. $W = Fd\cos 60^\circ = 57 \times 21 \times \frac{1}{2} = 1197 \times \frac{1}{2}$ J

11 [Intermediate] • Target: The Work-Energy Principle

A force of 38 N acts at 60° to the direction of motion of a body that moves 23 m. Find the work done, leaving your answer in exact (surd) form.

Final Answer: $874 \times \frac{1}{2}$ J. $W = Fd\cos 60^\circ = 38 \times 23 \times \frac{1}{2} = 874 \times \frac{1}{2}$ J

12 [Intermediate] • Target: Power and Vehicles in Motion

A force of 77 N acts at 60° to the direction of motion of a body that moves 7 m. Find the work done, leaving your answer in exact (surd) form.

Final Answer: $539 \times \frac{1}{2}$ J. $W = Fd\cos 60^\circ = 77 \times 7 \times \frac{1}{2} = 539 \times \frac{1}{2}$ J

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13 [Intermediate] • Target: Work Done by Constant & Inclined Forces

A body of mass 11 kg moving at 7 m/s has work 356 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

Final Answer: 10.66. $12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (7)^2 + 2(356)11 = 1251/11v10.66$ m/s

14 [Intermediate] • Target: Kinetic Energy & Potential Energy

A body of mass 11 kg moving at 1 m/s has work 789 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

Final Answer: 12.02. $12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (1)^2 + 2(789)11 = 1589/11v12.02$ m/s

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The Work-Energy Principle • Cross-Lesson: 10.01 Series & Binomial Theorem

A body of mass 7 kg moving at 11 m/s has work 470 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

Final Answer: 15.98. $12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (11)^2 + 2(470)7 = 1787/7v15.98$ m/s

16 [Challenging] • Target: Power and Vehicles in Motion • Cross-Lesson: 11.01 Kinematics

A body of mass 22 kg moving at 9 m/s has work 159 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

Final Answer: 9.77. $12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (9)^2 + 2(159)22 = 1050/11v9.77$ m/s

17 [Challenging] • Target: Work Done by Constant & Inclined Forces • Cross-Lesson: 12.01 Forces & Equilibrium

A body of mass 25 kg moving at 7 m/s has work 448 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

Final Answer: 9.21. $12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (7)^2 + 2(448)25 = 2121/25v9.21$ m/s

18 [Challenging] • Target: Kinetic Energy & Potential Energy • Cross-Lesson: 13.01 Newton's Laws

A body of mass 8 kg moving at 11 m/s has work 311 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

Final Answer: 14.10. $12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (11)^2 + 2(311)8 = 795/4v14.10$ m/s

19 [Challenging] • Target: The Work-Energy Principle • Cross-Lesson: 15.01 Representation of Data

A body of mass 8 kg moving at 8 m/s has work 195 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

Final Answer: 10.62. $12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (8)^2 + 2(195)8 = 451/4v10.62$ m/s

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20 [Challenging] • Target: Power and Vehicles in Motion • Cross-Lesson: 16.01 Permutations & Combinations

A body of mass 10 kg moving at 7 m/s has work 168 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

Final Answer: 9.09. $12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (7)^2 + 2(168)10 = 413/5v9.09$ m/s

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Work Done by Constant & Inclined Forces

A force of 60 N pulls a box across a horizontal floor through a distance of 8 m. The force acts at an angle of 60° above the horizontal. Find the work done by the force.

a. 240 J ✓

b. 480 J

c. 415.7 J

d. 120 J

Work done $W = Fd\cos\theta = 60 \times 8 \times \cos(60^\circ) = 480 \times 0.5 = 240$ J.

22 [Foundational] • Target: Work Done by Constant & Inclined Forces

A crate of mass 15 kg is lifted vertically at a constant speed through a height of 4 m. Taking $g = 10$ m/s^2, what is the work done against gravity?

a. 600 J ✓

b. 150 J

c. 300 J

d. 60 J

$W = mgh = 15(10)(4) = 600$ J.

23 [Foundational] • Target: Work Done by Constant & Inclined Forces

A box is pushed 5 m along a horizontal floor against a constant frictional resistance of 35 N. What is the work done against friction?

a. 175 J ✓

b. 150 J

c. 350 J

d. 70 J

$W = F_f \times d = 35 \times 5 = 175$ J.

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24 [Foundational] • Target: Kinetic Energy & Potential Energy

Calculate the kinetic energy of an athlete of mass 70 kg running at a speed of 8 m/s.

a. 2240 J ✓

b. 4480 J

c. 560 J

d. 280 J

$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(70)(64) = 35 \times 64 = 2240$ J.

25 [Foundational] • Target: Kinetic Energy & Potential Energy

A rollercoaster car of mass 500 kg starts from rest at the top of a frictionless hill of height 20 m. Taking $g = 10$ m/s^2, find its speed at the bottom of the hill.

a. 20 m/s ✓

b. 10 m/s

c. 40 m/s

d. 14.1 m/s

Conservation of energy: $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2(10)(20)} = \sqrt{400} = 20$ m/s.

26 [Foundational] • Target: Kinetic Energy & Potential Energy

A bullet of mass 0.02 kg traveling at 300 m/s penetrates a wooden block to a depth of 0.15 m before stopping. What is the average resistive force exerted by the wood?

a. 6000 N ✓

b. 3000 N

c. 9000 N

d. 12000 N

Work-energy principle: $Fd = \frac{1}{2}mv^2 \Rightarrow F(0.15) = \frac{1}{2}(0.02)(90000) = 900 \Rightarrow F = 900/0.15 = 6000$ N.

27 [Foundational] • Target: The Work-Energy Principle

A skier of mass 60 kg starts from rest at the top of a slope 30 m high. When reaching the bottom, her speed is 20 m/s. Taking $g = 10$ m/s^2, how much energy was lost due to friction?

a. 6000 J ✓

b. 12000 J

c. 18000 J

d. 3000 J

Initial potential energy $E_p = mgh = 60(10)(30) = 18000$ J. Final kinetic energy $E_k = \frac{1}{2}(60)(400) = 12000$ J. Energy lost $= 18000 - 12000 = 6000$ J.

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28 [Foundational] • Target: The Work-Energy Principle

A package of mass 4 kg slides down a rough incline of length 10 m and vertical drop 6 m. If the resistive force is a constant 8 N, find the speed of the package at the bottom (starting from rest, $g = 10$ m/s^2).

a. $\sqrt{80} \approx 8.94$ m/s ✓

b. 10 m/s

c. 8 m/s

d. 12 m/s

Work done by gravity $= mgh = 4(10)(6) = 240$ J. Work against friction $= 8 \times 10 = 80$ J. Net kinetic energy $= 240 - 80 = 160$ J. $\frac{1}{2}(4)v^2 = 160 \Rightarrow 2v^2 = 160 \Rightarrow v^2 = 80 \Rightarrow v = \sqrt{80} \approx 8.94$ m/s.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Work-Energy Principle

A particle is projected up a rough incline (30°) with initial speed 10 m/s. If $\mu = 0.2$ and $g = 10$ m/s^2, how far up the plane does the particle travel before coming to rest?

a. ≈ 7.43 m ✓

b. 10.0 m

c. 5.50 m

d. 8.25 m

Deceleration up slope: $a = g(\sin 30^\circ + \mu \cos 30^\circ) = 10(0.5 + 0.2 \times 0.866) = 10(0.5 + 0.1732) = 6.732$ m/s^2. Distance $s = u^2/(2a) = 100/(2 \times 6.732) \approx 7.43$ m.

30 [Intermediate] • Target: Power and Vehicles in Motion

A car of mass 1200 kg travels along a level road at a constant speed of 30 m/s. The engine is working at 45 kW. What is the total resistance to motion?

a. 1500 N ✓

b. 1200 N

c. 1350 N

d. 2000 N

$P = Fv \Rightarrow 45000 = F(30) \Rightarrow F = 1500$ N. Since speed is constant, driving force equals resistance: $R = 1500$ N.

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31 [Intermediate] • Target: Power and Vehicles in Motion

The same car (mass 1200 kg, engine working at 45 kW) faces the same resistance of 1500 N while traveling at 20 m/s. What is its acceleration at this instant?

a. $0.625\ \text{m/s}^2$ ✓

b. $0.75\ \text{m/s}^2$

c. $1.00\ \text{m/s}^2$

d. $0.50\ \text{m/s}^2$

Driving force at 20 m/s: $F = P/v = 45000/20 = 2250$ N. Net force $= 2250 - 1500 = 750$ N. Acceleration $a = 750/1200 = 0.625\ \text{m/s}^2$.

32 [Intermediate] • Target: Power and Vehicles in Motion

A truck of mass 5000 kg ascends an incline of 1 in 20 ($\sin\alpha = 1/20$) at a steady speed of 15 m/s. The engine power is 60 kW. Taking $g = 10\ \text{m/s}^2$, find the resistance to motion.

a. 1500 N ✓

b. 2500 N

c. 4000 N

d. 2000 N

Driving force $F = P/v = 60000/15 = 4000$ N. Gravity component along slope $= mg\sin\alpha = 5000(10)(1/20) = 2500$ N. At steady speed:
$F = mg\sin\alpha + R \Rightarrow 4000 = 2500 + R \Rightarrow R = 1500$ N.

33 [Intermediate] • Target: Work Done by Constant & Inclined Forces

A body of mass 10 kg moving at 4 m/s has work 124 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 5.39

b. **6.39** ✓

c. 9.58

d. 7.39

$12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (4)^2 + 2(124)10 = 204/5v6.39$ m/s

34 [Intermediate] • Target: Kinetic Energy & Potential Energy

A body of mass 3 kg moving at 7 m/s has work 218 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. **13.94** ✓

b. 14.94

c. 20.91

d. 12.94

$12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (7)^2 + 2(218)3 = 583/3v13.94$ m/s

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35 [Intermediate] • Target: The Work-Energy Principle

A body of mass 7 kg moving at 5 m/s has work 316 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. **10.74** ✓

b. 16.11

c. 11.74

d. 9.74

$12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (5)^2 + 2(316)7 = 807/7v10.74$ m/s

36 [Intermediate] • Target: Power and Vehicles in Motion

A body of mass 15 kg moving at 7 m/s has work 305 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 14.20

b. 10.47

c. 8.47

d. **9.47** ✓

$12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (7)^2 + 2(305)15 = 269/3v9.47$ m/s

37 [Intermediate] • Target: Work Done by Constant & Inclined Forces

A body of mass 24 kg moving at 0 m/s has work 544 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 5.73

b. **6.73** ✓

c. 10.10

d. 7.73

$12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (0)^2 + 2(544)24 = 136/3v6.73$ m/s

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Kinetic Energy & Potential Energy • Cross-Lesson: 13.01 Newton's Laws

A body of mass 6 kg moving at 0 m/s has work 292 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. **9.87** ✓

b. 14.80

c. 10.87

d. 8.87

$12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (0)^2 + 2(292)6 = 292/3v9.87$ m/s

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39 [Challenging] • Target: The Work-Energy Principle • Cross-Lesson: 15.01 Representation of Data

A body of mass 7 kg moving at 0 m/s has work 134 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 6.19 ✓

b. 7.19

c. 5.19

d. 9.28

$12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (0)^2 + 2(134)7 = 268/7v6.19$ m/s

40 [Challenging] • Target: Power and Vehicles in Motion • Cross-Lesson: 16.01 Permutations & Combinations

A body of mass 7 kg moving at 8 m/s has work 45 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 9.77

b. 8.77 ✓

c. 13.15

d. 7.77

$12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (8)^2 + 2(45)7 = 538/7v8.77$ m/s

41 [Challenging] • Target: Work Done by Constant & Inclined Forces • Cross-Lesson: 17.01 Discrete Random Variables

A body of mass 6 kg moving at 8 m/s has work 186 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 11.22 ✓

b. 12.22

c. 10.22

d. 16.84

$12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (8)^2 + 2(186)6 = 126v11.22$ m/s

42 [Challenging] • Target: Kinetic Energy & Potential Energy • Cross-Lesson: 18.01 Binomial Distribution

A body of mass 3 kg moving at 9 m/s has work 282 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 16.40 ✓

b. 24.60

c. 17.40

d. 15.40

$12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (9)^2 + 2(282)3 = 269v16.40$ m/s

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43 [Challenging] • Target: The Work-Energy Principle • Cross-Lesson: 19.01 Normal Distribution

A body of mass 6 kg moving at 7 m/s has work 76 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 9.62

b. 7.62

c. **8.62** ✓

d. 12.93

$12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (7)^2 + 2(76)6 = 223/3v8.62$ m/s

44 [Challenging] • Target: Power and Vehicles in Motion • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A body of mass 2 kg moving at 7 m/s has work 99 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 18.25

b. **12.17** ✓

c. 11.17

d. 13.17

$12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (7)^2 + 2(99)2 = 148v12.17$ m/s

45 [Challenging] • Target: Work Done by Constant & Inclined Forces • Cross-Lesson: 3.01 Graphs & Transformations

A body of mass 10 kg moving at 7 m/s has work 200 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

a. 10.43

b. 14.15

c. 8.43

d. **9.43** ✓

$12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (7)^2 + 2(200)10 = 89v9.43$ m/s

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